

James Sullivan

Senior Software Engineer

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SUMMARY/OBJECTIVE

Experienced Senior Software Engineer with over 13 years of expertise in developing high-performance, scalable applications across various domains. Proven track record in leading performance optimization initiatives, cloud-native solutions, and seamless system migrations. Expert in designing and optimizing complex SQL queries, database schemas, and RESTful APIs, with hands-on experience in **Python, Java, React, Node, Golang, MongoDB, PostgreSQL**, and Azure. Skilled in leveraging modern tools and frameworks such as **FastAPI, Django, Flask, React, Kubernetes**, and AWS to deliver efficient, reliable, and scalable software solutions. Adept at driving cross-functional collaboration, automating CI/CD pipelines, and ensuring high-quality software delivery.

EXPERIENCE

Acrisure

Sr. Software Engineer

Plant City, Florida

Oct 2020 – Jan 2025

- Spearheaded the design and development of **RESTful APIs** using **Flask**, enhancing microservice communication and improving data flow and system architecture.
- Refactored and optimized **Django** views and database queries into **FastAPI** endpoints, ensuring seamless user experience while maintaining data model compatibility.
- Designed and deployed **RESTful APIs** with **Python** and **Azure API Management**, ensuring secure, scalable interactions between microservices and external clients with high availability and efficiency.
- Leveraged **Django Ninja's** dependency injection to improve code modularity and testing, facilitating easier maintenance and more efficient unit testing of API endpoints.
- Designed and developed high-performance, low-latency APIs in **Golang** for distributed systems, resulting in improved request throughput and system responsiveness.
- Built and optimized real-time APIs for serving **machine learning** model predictions with **Flask** and **FastAPI**, enhancing decision-making capabilities in production environments.
- Led the migration of legacy **Django** backend services to **FastAPI**, reducing request latency and improving overall response times by leveraging FastAPI's **asynchronous** capabilities.
- Implemented modern **asynchronous Python** features in **FastAPI** to optimize I/O-bound operations such as API calls and database interactions, significantly boosting throughput.
- Refactored legacy Python code to enhance readability, maintainability, and scalability, adhering to **PEP 8** standards and best practices.
- Identified and resolved performance bottlenecks in **Python** applications by optimizing algorithms and utilizing efficient data structures.
- Rewrote unreliable and poorly documented **Python** code to improve reliability and testability, incorporating comprehensive unit tests and adopting **Test-Driven Development (TDD)**.
- Modularized a monolithic codebase into reusable **Python** modules, improving code organization and simplifying onboarding for new developers.
- Rewrote complex and unstructured **Python** scripts into more efficient, maintainable solutions, ensuring clarity with detailed code comments and adhering to software engineering best practices.
- Optimized **database queries** in **Python** applications by implementing efficient **ORM** practices with **SQLAlchemy** or **Django ORM**, reducing database load and improving response times.
- Streamlined **Python**-based data pipelines, simplifying error handling and ensuring fault tolerance in **ETL** processes for better performance.

- Integrated **Elixir-based microservices** with **Python** applications via **RESTful APIs** and message brokers like **RabbitMQ**, enhancing system scalability and fault tolerance.
- Developed and maintained **Python**-based data processing workflows using **Elixir's** concurrency model, optimizing processing times for large datasets.
- Led performance optimization initiatives for Python applications, focusing on identifying and resolving bottlenecks in **database queries** and **memory usage**, resulting in significant improvements in application speed and performance.
- Designed and optimized complex **SQL** queries and stored procedures, improving data processing workflows and reducing query execution times for more efficient data retrieval.
- Directed the migration from **SQL**-based databases to **MongoDB**, leveraging MongoDB's flexible schema and indexing features to accelerate data retrieval for large datasets.
- Enhanced **MongoDB** performance by optimizing replica set and sharding configurations, increasing availability and scalability, and minimizing downtime during production incidents.
- Developed and maintained Python applications using **SQLAlchemy** and **SQLModel** for efficient PostgreSQL database interactions, ensuring optimized query execution and data retrieval.
- Designed and optimized database schemas with **SQLAlchemy ORM**, reducing complex joins and improving query performance in PostgreSQL.
- Collaborated with cross-functional teams to optimize cloud-based **Python** applications on **AWS**, utilizing **EC2**, **S3**, and **Lambda** services to reduce infrastructure costs through efficient resource management.
- Configured and deployed applications on **Amazon Elastic Kubernetes Service (EKS)**, leveraging **Kubernetes** for container orchestration, ensuring seamless scaling and high availability of services.
- Developed a comprehensive **containerization strategy** using **Docker** and **Kubernetes**, enabling seamless deployment and scaling of **Python** microservices across multi-cloud environments.
- Worked closely with development teams to integrate **Python**-based services into **Kubernetes** clusters, addressing resource allocation, networking, and logging challenges.
- Optimized **CI/CD** pipelines for **Python** applications using **GitLab CI**, **Jenkins**, and **GitHub Actions**, streamlining the release process and improving development cycles while maintaining high-quality standards.
- Led the creation of automated **CI/CD** pipelines with **Docker**, **Jenkins**, and **Kubernetes**, improving release cycles by automating builds, testing, and deployment, ensuring high-quality software delivery.
- Integrated **NGINX** with **Python**-based applications (**Flask/Django**) for reverse proxying, caching, and session management, reducing response times and improving overall performance by offloading static content handling.
- Integrated **PyTest** into **CI** pipelines using **Jenkins** and **GitLab CI**, automating test execution, code quality checks, and report generation, which accelerated release cycles and minimized manual testing efforts.
- Managed end-to-end deployment pipelines using **Ansible**, simplifying continuous delivery and reducing deployment times while improving release predictability for **Python** applications.
- Optimized **CircleCI** workflows by implementing caching and parallel job execution strategies, reducing build times and speeding up overall **CI/CD** processes for **Python** applications.
- Collaborated with product and engineering teams to define **JIRA** epics, stories, and tasks, ensuring clear project scope and requirements, enhancing sprint planning, and reducing scope creep.
- Worked with cross-functional teams to integrate **Mendix-based** applications with **Python**-driven analytics tools, enabling real-time data analysis and reporting, and improving data-driven decision-making.
- Led the development of full-stack applications using **React** (front-end) and **Node.js** (back-end), improving feature delivery and team efficiency by adopting a unified **JavaScript** stack.
- Contributed to the development of a real-time **analytics platform** for engineering teams using **Plotly.js** and **WebSocket** technology, allowing dynamic updates and visualizations for time-series data.
- Optimized **Python** applications for performance on **Linux** systems, using tools like **psutil**, **pdb**, and profiling techniques to analyze resource consumption and speed.
- Leveraged **Linux cron jobs** to schedule **Python** scripts for routine system maintenance tasks, such as log rotation and data synchronization.
- Collaborated with UI/UX designers using **Figma** to create wireframes and interactive prototypes for web applications, ensuring intuitive and visually appealing user interfaces.
- Configured **WebSockets** in **Node.js** applications for real-time communication between client and server, enhancing user engagement in collaborative applications.
- Implemented secure **WebSocket** connections using **NGINX** as a reverse proxy, ensuring scalability and performance.

- Provided real-time feedback and revisions to improve user flows and interface consistency across multiple devices.
- Integrated **SAP ECC** with external web applications using **SAP PI/PO**, enabling seamless data exchange and process automation.
- Developed and deployed custom **SAP Fiori** applications to enhance the user interface and interaction with SAP back-end systems.
- Optimized **Java-based backend** systems for performance and scalability, using **multithreading** and Java's **CompletableFuture** to improve concurrency, reducing response times and enhancing throughput for high-traffic endpoints.
- Developed and automated build and deployment pipelines using **Shell scripts**, integrating **.NET applications** with **CI/CD tools** like **Jenkins** and **Azure DevOps** for continuous integration and delivery.
- Created **Shell scripts** to automate **.NET environment configurations**, improving consistency and efficiency in development, testing, and production deployments.
- Implemented **Alembic** for database schema management in complex multi-database environments, automating schema versioning and rollback procedures, reducing errors during updates, and simplifying deployment processes.

Spot Freight

Senior Software Developer

Tampa, Florida

May 2017 – May 2020

- Implemented **Tailwind CSS** for creating responsive and visually appealing UI components in **Python** web applications, minimizing custom CSS development time and ensuring consistent styling across the application.
- Created state-driven **Vue.js** components integrated with Python **Flask/Django** APIs to enable real-time data updates, enhancing UI interactivity and improving overall application performance.
- Built lightweight, scalable **single-page applications (SPAs)** using **Vue.js** in Python development environments, delivering a modern, reactive user interface and optimizing user experience.
- Integrated **React** and **Node.js** into an existing Python-based backend, enhancing the user experience through dynamic front-end updates and enabling smooth, real-time data exchanges via **RESTful APIs** and **GraphQL** endpoints.
- Streamlined **CRUD** operations by integrating **Java Jersey** with **PostgreSQL**, leveraging Jersey's built-in support for data serialization and deserialization to enhance database interaction efficiency.
- Optimized microservices architecture by implementing **Gunicorn** as the **WSGI** HTTP server for Python web applications, fine-tuning worker configurations to handle varying traffic loads and reducing latency during peak usage times.
- Designed and implemented high-performance **PostgreSQL** database architectures, improving query performance through advanced indexing strategies and ensuring data integrity for applications with complex data models.
- Integrated complex business logic into the **ORM layer** using **SQLAlchemy**, enabling efficient data validation and seamless interaction between Python code and **PostgreSQL**, while leveraging declarative mapping for enhanced maintainability.
- Automated UI testing using **Selenium WebDriver** to perform end-to-end testing of Python-based web applications, ensuring high-quality user experiences through functional and regression testing.
- Integrated **Selenium** with Python for browser automation and functional tests, reducing manual testing efforts and enhancing the reliability and performance of web applications.
- Developed and maintained UI test suites with **Jest**, validating JavaScript behavior within the Python-driven back-end environment, ensuring seamless interaction between the front-end and back-end layers.
- Implemented **Playwright** for Python to automate browser tests for modern web applications, improving test reliability and performance with headless browser support.
- Used **Playwright** to simulate real-world user interactions with Python back-end services, testing complex workflows such as form submissions, API calls, and dynamic content loading.
- Designed and built a custom **Buildkite** agent to automate deployment pipelines, optimizing build times through parallelism and better resource allocation, resulting in faster and more efficient deployment cycles.

- Created end-to-end **CI/CD** pipelines with **Jenkins**, **GitLab CI**, and **CircleCI**, automating the build, test, and deployment workflows to reduce release times and improve overall development cycle efficiency.
- Architected and developed **GraphQL APIs** with Python (**Graphene**), enabling flexible and efficient data querying and enhancing frontend performance by supporting real-time, dynamic data fetching based on user interactions.
- Introduced **Confluence** as the central knowledge-sharing platform, creating and maintaining technical documentation, release notes, and team guides, which improved cross-team communication and accelerated onboarding for new developers.
- Automated database migrations and schema changes using **Alembic**, ensuring smooth updates to the **PostgreSQL** schema and maintaining consistency across different environments.
- Developed cloud-native applications on **AWS** utilizing **EC2**, **S3**, **RDS**, and **Lambda** to create scalable and high-performance systems, improving application availability and significantly reducing infrastructure costs

Timspark

Senior Software Developer

St. Petersburg, Florida

Feb 2014 – Feb 2017

- **Led** front-end development for an application, utilizing **Python** for scripting and **Django** to provide robust back-end support, ensuring seamless integration and performance.
- Created a high-performance **API** with **Python**, optimizing it for handling a large volume of requests with minimal latency.
- **Contributed** to the development of a healthcare application using **Angular**, **HTML**, and **CSS** to create an intuitive and accessible user interface, enhancing patient interaction and care delivery.
- **Performed** data preprocessing, including handling missing values, encoding categorical variables, and scaling numerical features to prepare datasets for model training.
- Developed and optimized **machine learning** models, such as logistic regression, decision tree, and random forest, to identify the highest-performing solution.
- Utilized **Python** and **JavaScript** to build reusable, scalable components across the application framework, ensuring consistency and efficiency in development.
- Executed complex database queries using **Python-MySQL connector** and **MySQL**, retrieving essential data to support application functionality.
- Designed an **API** following **OpenAPI standards**, providing clear documentation and facilitating easy client integration for seamless data access.
- Designed and created databases using **MySQL** and wrote efficient queries for data extraction, enabling smooth data operations within the application.
- Worked with the backend team to design and implement **RESTful services** using **Node.js**, improving data interactions within a **Vue.js** application.
- Actively participated in **code reviews** and pair programming sessions, fostering a collaborative culture of continuous learning, knowledge sharing, and code quality.

EDUCATION

Bachelor Degree of **Computer Science** at Florida State University -2013

STRENGTH

Goal-oriented, Enthusiastic, Quick adapting, Self-motivated, Team collaboration, Communication skills